DOTUN OLAOYE

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SUMMARY

PhD student in Statistical Genetics specializing in quantitative genetics and genomic prediction across animal and plant breeding systems. Experienced in genome-phenome association analysis, trait prediction, and mixed-model genetic evaluation using both frequentist and Bayesian statistical frameworks. Background in plant pathology and disease-resistance breeding, with three years of industry experience evaluating crop disease resistance within a commercial trait-advancement pipeline.

EDUCATION

Doctor of Philosophy - Statistical Genetics (with Dr. Jared Decker)
University of Missouri, Columbia. June 2024 - Present
Research focus: multi-breed genomic prediction of puberty and fertility traits in beef heifer cattle

Master of Science - Plant Breeding, Genetics and Pathology
University of Arkansas, Fayetteville. May 2021

Bachelor of Agriculture (Honor's: Crop Protection/Pathology)
Federal University of Agriculture, Abeokuta, Nigeria. 2017

WORK EXPERIENCE

Graduate Research Assistant, University of Missouri, Columbia. June 2024 - Present

- Built and validated a multi-breed genomic prediction pipeline (univariate, bivariate, and multi-trait single-step GBLUP and GBLUP) for puberty and fertility traits across five beef cattle breed populations (Angus, Red Angus, Hereford, Bos indicus-influenced, and a pooled MultiBreed population of over 9,000 animals genotyped on an 800K SNP panel) using BLUPF90+/GIBBSF90+
- Estimated variance components, heritability, and genetic correlations via REML/AI-REML, and validated genomic prediction accuracy using a Linear Regression (LR) temporal validation framework
- Presented genomic prediction research at the Advances in Genome Biology and Technology (AGBT) Agricultural Meeting, 2025, and the Population, Evolutionary, and Quantitative Genetics Conference (PEQG), 2026

Collaborative Research Project (ongoing, independent of PhD assistantship duties, originated during MS), University of Arkansas collaborators

- Extended weighted genomic prediction methodology to disease-resistance and abiotic-stress trait prediction in cowpea and common bean, using SNP weighting informed by meta-transcriptomic (RNA-seq) evidence

Pathology Research Specialist, Pathology Unit, Trait Assessment, North America, Syngenta Seeds Inc. May 2021 - April 2024

- Assessed field corn, sweet corn, and soybean for resistance or tolerance to plant pathogens of economic importance, feeding disease trait data directly into product advancement decisions
- Planned and implemented pathology research trials on field corn within the Trait Assessment program
- Served as a member of Market Segment Teams for Field Corn pathology traits in North America
- Handled the adoption of a proprietary data collection tool for field corn pathology trials

Program Technician, Vegetable Breeding and Genetic Laboratory, Department of Horticulture, University of Arkansas, Fayetteville. April 2021 - May 2021

- Worked on improving economically important traits in spinach, arugula, soybean, and common bean

Graduate Research Assistant, spinach breeding and genetics for downy mildew resistance, University of Arkansas, Fayetteville. January 2019 - April 2021

- Bred spinach for resistance to the downy mildew pathogen *Peronospora effusa*, including genetic characterization of resistance loci and race-specific resistance across worldwide germplasm
- Conducted disease assays and phenotyping for disease and agronomic traits in field corn, sweet corn, soybean, and snap bean, in field and greenhouse settings
- Produced high-quality inoculum for disease assays and standardized disease assay protocols

Laboratory Attendant, Plant Pathology Unit, International Institute of Tropical Agriculture, IITA-CGIAR, Ibadan, Nigeria. November 2018 - January 2019

- Biocontrol management strategies for black Sigatoka disease of banana plants

Graduate Research Unpaid Internship, Plant Pathology Unit, International Institute of Tropical Agriculture, IITA-CGIAR, Ibadan, Nigeria. June 2018 - November 2018

- Characterized the biology of plant pathogens causing diseases of economic importance in field corn and banana

Undergraduate Research Assistantship, Department of Crop Protection, Federal University of Agriculture, Abeokuta, Nigeria. June 2015 - October 2016

- Participated in molecular marker evaluation of indigenous tomato genotypes for Fusarium wilt resistance (CAPS marker TAO1, endonuclease FOK1, homozygous I2 resistance gene selection)

LEADERSHIP

Genetics Students' Representative, Graduate Professional Council, University of Missouri-Columbia

Trained new hires and supervised seasonal technical personnel and a research associate within the Field Corn Pathology team, Trait Assessment, North America, Syngenta Seeds Inc.

Currently mentoring an honors student in the research lab

Mentored a bachelor's student on manuscript preparation and publication (see acknowledgement: https://doi.org/10.1080/03235408.2020.1715756)

PROFESSIONAL SERVICE

Served as Ad-hoc Secretary for the Vigna spp USDA's CGC at the Annual International Crop Science Meeting, Texas, USA. November 2019

Judge, Agronomy and Horticulture 4-H program, University of Minnesota. July 2022

Invited as a panelist for the 2023 Professional Development Workshop, Corteva Symposium, West Africa

Volunteer Reviewer: Agrosystems, Geosciences & Environment; Nigerian Journal of Biotechnology

- Provided reviews on three manuscripts for the journal, Nigerian Journal of Biotechnology
- Provided reviews on a few manuscripts for the journal, Agrosystems, Geosciences & Environment

Reviewed at least 10 conference abstracts for Plant Health 2024, American Phytopathological Society

Member and volunteer, African Student Organization, University of Arkansas, Fayetteville. 2019-2021

Volunteer member, Volunteer Action Center, Fayetteville, Arkansas. 2019-2021

KEY SKILLS

Quantitative genetics / genomic prediction: Single-step GBLUP (ssGBLUP) and GBLUP via BLUPF90+/GIBBSF90+, including threshold models for binary traits; univariate, bivariate, and multi-trait mixed-model analyses; both frequentist (REML/AI-REML) and Bayesian (Gibbs sampling/MCMC via GIBBSF90+) variance component estimation; Linear Regression (LR) validation; genomic relationship matrix construction (VanRaden method); weighted/evidence-informed genomic prediction

Population genetics: Principal component analysis and population stratification correction (PLINK); genotype imputation (Beagle); variant filtering and manipulation (bcftools)

Programming: R, Python, Perl, SQL, SAS, JMP, Quarto/R Markdown, shell scripting, HPC job orchestration (SLURM)

Genomic/bioinformatics tools: BLUPF90+/GIBBSF90+ family, PLINK, bcftools, Beagle, TASSEL, MEGA, QTL mapping and GWAS software

Laboratory and field: Inoculum production for disease assays, disease assay standardization, phenotypic data collection, marker-trait association analysis, tissue collection for genomic and phenotypic analysis

AWARDS

FFAR Fellow (Stipend + Professional Development; ~$148,500 over 3 years: student stipend, tuition, and fees), Foundation for Food and Agriculture Research, in partnership with North Carolina State University. 2025-2028

- Highly competitive fellowship designated for high-potential scientists, funding 150+ hours of professional development and leadership training
- Training activities included Professional Development Plans, Mentoring, Project Management, Designing and Managing Meetings, Facilitating Participatory Decision-Making, Conflict Management, Cross-Cultural Dynamics, 360-degree Leadership, Effective Mentoring, Crucial Conversations and Crucial Influencing, Productivity Workshop, Effective Science Communication, and Policy Communication, and a visit to the United States Congress to advocate for agricultural research funding

2025 Scholars of Distinction, University of Missouri Graduate School (recognizing external funding recipients). 2025

PhD Graduate Research Assistantship, Genetics Area Program, University of Missouri, Columbia. 2024 - Present

MS Graduate Research Assistantship (~$66,830), Department of Horticulture, University of Arkansas, Fayetteville. 2019-2021 (Funded by a USDA grant to work on a Specialty Crop)

2020 Conference Travel Grant Award ($350), American Society for Horticultural Science

Two-time (2019 & 2020) Travel Grant Award ($600), Graduate School, University of Arkansas, Fayetteville

2019 Plant Health and Quality Summer School Scholarship (€1200), Campus Vegetal of Angers, University of Angers, France

PUBLICATIONS

Olaoye, D., Bhattarai, G., Correll, J., Shi, A. Genetic analysis reveals novel genomic regions for downy mildew resistance and validates the potential for predictive breeding in spinach (*Spinacia oleracea*). Under review, BMC Plant Biology, 2026.

Olaoye, D., Rasaki, L., Adesina, O., Kareem, B., Kandel, S., Ravelombola, W., Yang, Y., Shi, A. RNA-seq meta-analysis and machine learning identify stress-responsive genes and improve genomic prediction in common bean (*Phaseolus vulgaris* L.) with cross-species application in cowpea (*Vigna unguiculata* L.). bioRxiv (2026). https://doi.org/10.64898/2026.08.08.743654 (preprint)

Olaoye, D., Bhattarai, G., Feng, C. et al. Evaluation of downy mildew resistance in spinach (*Spinacia oleracea*). Euphytica 220, 38 (2024). https://doi.org/10.1007/s10681-023-03289-9

Bhattarai G, **Olaoye D**, Mou B, Correll JC and Shi A (2022) Mapping and selection of downy mildew resistance in spinach cv. whale by low coverage whole genome sequencing. Front. Plant Sci. 13:1012923. doi: 10.3389/fpls.2022.1012923

Zia, B, A. Shi, **D. Olaoye**, H. Xiong, W. Ravelombola, P. Gepts, H.F. Schwartz, M.A. Brick, K. Otto, B. Ogg, and S, Chen. 2022. Genome-wide association study and genomic prediction for bacterial wilt resistance in common bean (*Phaseolus vulgaris*) core collection. Front. Genet. https://doi.org/10.3389/fgene.2022.853114

Ravelombola, W., Shi, A., Huynh, B.L., Qin, J., Xiong, H., Manley, A., Dong, L., **Olaoye, D.**, Bhattarai, G., Zia, B. and Alshaya, H., 2022. Genetic architecture of salt tolerance in a Multi-Parent Advanced Generation Inter-Cross (MAGIC) cowpea population. BMC genomics, 23(1), pp.1-22.

Ravelombola, W., Dong, L., Barickman, T.C., Xiong, H., **Olaoye, D.**, Bhattarai, G., Zia, B., Alshaya, H., Alatawi, I. and Shi, A., 2021. Evaluation of salt tolerance in cowpea at seedling stage. Euphytica, 217(6), pp.1-20.

Ravelombola, W., Shi, A., Chen, S., Xiong, H., Yang, Y., Cui, Q., **Olaoye, D.** and Mou, B., (2020). Evaluation of cowpea for drought tolerance at seedling stage. Euphytica, 216(8), pp.1-19.

Yang, Y., Shi, D., Wang, Y., Zhang, L., Chen, X., Yang, X. ... & Yang, G. (2020). Transcript profiling for regulation of sweet potato skin color in Sushu8 and its mutant Zhengshu20. Plant Physiology and Biochemistry, 148, 1-9.

PRESENTATIONS

Presenter (poster): Olaoye, D., Franklin, G., Schnabel, R., Thomas, J., Decker, J. Predicting Puberty and Fertility Traits in Multi-breed Beef Heifer Cattle. Population, Evolutionary, and Quantitative Genetics Conference (PEQG), 2026

Co-author (poster): Franklin, G., Olaoye, D., Thomas, J., Decker, J. Genome-wide association studies of puberty and fertility traits in US beef heifers. Population, Evolutionary, and Quantitative Genetics Conference (PEQG), 2026

Presenter: Estimating genetic parameters and predicting puberty and fertility traits in multi-breed beef heifer cattle. Advances in Genome Biology and Technology (AGBT) Agricultural Meeting, 03/31-04/2, 2025

Presenter: Olaoye, D., Bhattarai, G., Feng, C., Correll, J. C., & Shi, A. (2020, August). Evaluation of Resistance to Pfs race 5 in Worldwide Spinach Collection. In 2020 ASHS Annual Conference. ASHS.

Presenter: Olaoye, D., Shi, A., Ravelombola, W. S., Bhattarai, G., & Xiong, H. (2020, August). Drought Tolerance Characterization in two Contrasting Common Bean Genotypes. In 2020 ASHS Annual Conference. ASHS.

Presenter: Olaoye, D., Bhattarai, G., Feng, C., Shi, A., & Correll, J. (2019, November). Evaluation of Incomplete Dominance in a Spinach Downy Mildew Resistance Locus. In ASA, CSSA and SSSA International Annual Meetings (2019).

Co-author: Yang, Y., Cui, Q., Qin, J., Ravelombola, W., Bhattarai, G., Zia, B., ... & Shi, A. (2019, November). Evaluation of Drought Tolerance in Common Bean at Seedling Stage. In ASA, CSSA and SSSA International Annual Meetings (2019).

Co-author: Effect of Exogenous Application of Plant Growth Regulator in the Control of *Fusarium oxysporum* f. sp. *lycopersici* using two Tomato Genotypes. International Conference of the Biotechnology Society of Nigeria, 2018

Co-author: CAPS Marker TAO1 and Endonuclease FOK1 in the Selection of Tomato Genotypes with Homozygous I2 Resistance Gene to Fusarium Wilt. International Conference of the Biotechnology Society of Nigeria, 2018

PROFESSIONAL MEMBERSHIP AND TRAINING

Current Graduate Student Member: Genetics Society of America (GSA)

Former Graduate Student Member: American Society of Horticultural Science; American Society of Agronomy; Crop Science Society of America; Soil Science Society of America

Bruce Weir Summer Institute in Statistical Genetics (SISG). Summer 2025
Modules completed: QG1 (Quantitative Genetics), QG2 (Mixed Models for Genetic Analysis), HE3 (Artificial Intelligence/Machine Learning)

5th Annual Meeting of the Arkansas Bioinformatics Consortium (AR-BIC 2019). Bioinformatics in Food and Agriculture, UAMS, Little Rock, AR. February 25-26, 2019

Molecular biology workshop, Inqaba Biotec West Africa, University of Ibadan. April 2018